
Optical flow

Sofía Lucía - up202200682

IMCV- FEUP

Input sequences characteristics



- MPI Sintel Flow Dataset
- Name: temple-2
- Frames: 49
- Dimensions: (436, 1024, 3)

Farneback Algorithm

1. Image Pyramids

This pyramid helps the algorithm to capture motion at different scales.

2. Motion Estimation

By comparing the corresponding pixels in consecutive frames. It computes a polynomial expansion of the motion field using the Taylor series.

3. Local Polynomial Expansion

To model the local motion around each pixel. The polynomial expansion involves terms up to a certain order, and the coefficients of these terms are computed through the use of spatial and temporal gradients.

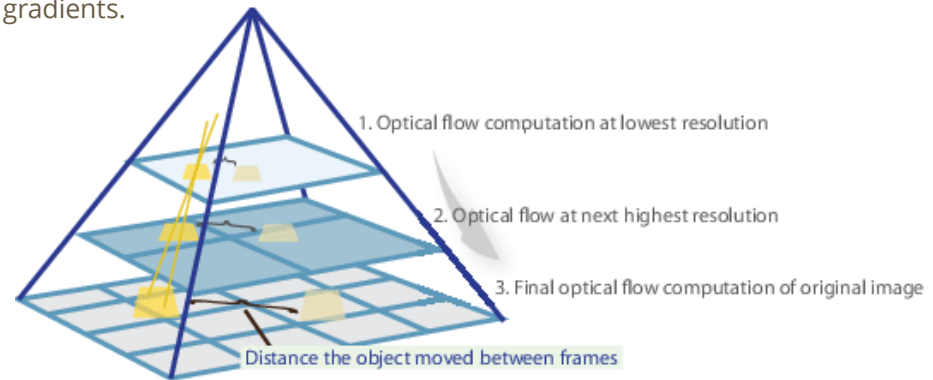
4. Gaussian Smoothing

Gaussian smoothing is applied to the images and the gradients before estimating the flow.

5. Iterative Refinement

At each iteration, it refines the flow field using the local polynomial expansion.

6. Final Flow Field



Farneback Output

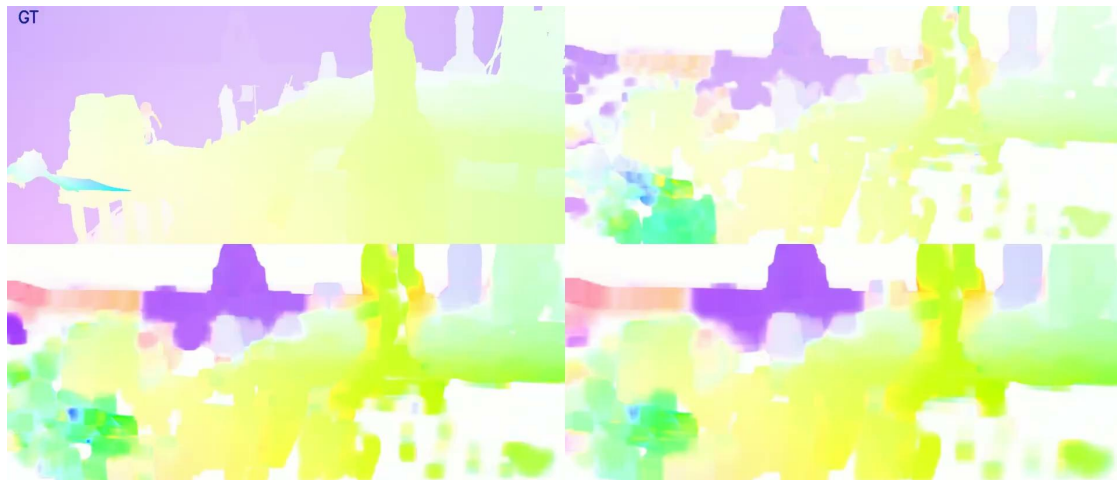


Vid 1. Groundtruth Vs Estimated with default parameters

MRSE	MAE	EPE	Time (s)
14.94	7.86	7.86	0.099

pyr_scale=0.5, levels=3, winsize=15,
iterations=3, poly_n=5, poly_sigma=1.1

Farneback Output - Testing different parameters



Bottom Left

pyr_scale=0.3, levels=5, **window_size=25**,
iterations=3, poly_n=5, poly_sigma=0.2

MRSE	MAE	EPE	Time (s)
14.15	7.00	7.00	0.077

Top Right

pyr_scale=0.3, levels=5, **window_size=15**,
iterations=3, poly_n=5, poly_sigma=0.2

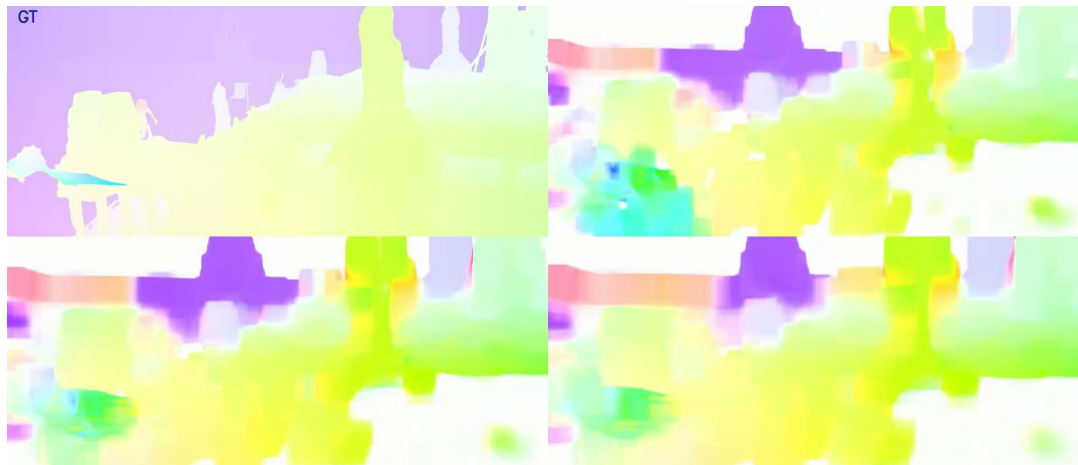
MRSE	MAE	EPE	Time (s)
14.33	7.23	7.23	0.081

Bottom Right

pyr_scale=0.3, levels=5, **window_size=35**,
iterations=3, poly_n=5, poly_sigma=0.2

MRSE	MAE	EPE	Time (s)
14.08	6.96	6.97	0.079

Farneback Output - Testing different parameters



Top Right
pyr_scale=0.3, levels=2, **window size=35**,
iterations=3, poly_n=5, poly_sigma=0.8

MRSE	MAE	EPE	Time (s)
14.08	7.05	7.05	0.069

Bottom Left
pyr_scale=0.3, levels=2, **window size=45**,
iterations=3, poly_n=5, poly_sigma=0.8

MRSE	MAE	EPE	Time (s)
14.02	7.02	7.02	0.069

Bottom Right
pyr_scale=0.3, levels=2, **window size=55**,
iterations=3, poly_n=5, poly_sigma=0.8

MRSE	MAE	EPE	Time (s)
13.98	7.02	7.02	0.069

RLOF Algorithm

The RLOF implementation can be seen as an improved pyramidal iterative Lucas-Kanade, which are:

- A **more robust redecending M-estimator framework** to improve the accuracy at motion boundaries and appearing and disappearing pixels.
- An **adaptive support region strategies** to reduce the corona effect.
- To **deal with illumination changes**, the intensity constancy assumption has been adopt with the Gennert and Negahdaripour illumination model.
- By using a **global motion prior initialization of the iterative refinement** the accuracy could be significantly improved for large displacements.
- **Allows** to compute dense optical flow fields in **real-time**.

RLOF Output



Top-right video
Params by default (gridStep =
Size(6, 6))

Bottom-left video
gridStep = Size(4, 4)

Bottom-right video
gridStep = Size(12, 12)

Comparative

	MRSE	MAE	EPE	Time (sec/frame)
Farneback (default)	14.94	7.86	7.86	0.099
Farneback (l=5, ws=35, it=3, pn=5, ps=0.2)	14.08	6.96	6.97	0.079
Farneback (l=2, ws=55, it=3, pn=3, ps=0.8)	13.98	7.02	7.02	0.069
RLOF (gridStep=(6,6))	11.3	4.67	4.67	0.38
RLOF (gridStep=(4,4))	10.96	4.35	4.35	0.73
RLOF (gridStep=(12,12))	12.47	4.77	5.77	0.16

Thank you!